Math Owl: what the app is and who it is for

In short

Math Owl, "The island of math whizzes", is an educational math game that children play in the browser, on a computer or a phone. With it, children learn and practise **multiplication, division, addition and subtraction** and prepare for the **Romanian National Assessment at the end of 2nd grade**.

The exercises follow the **Romanian school textbook**: the "Discover – Apply – Practice – Solve problems" lessons, the "Remember" boxes and the problems from the book. Everything is presented as an adventure on a map of islands.

The app is available in **Romanian and English**. The language can be changed at any time from the top bar (RO RO / GB EN).

Who it is for

- **Children in 2nd and 3rd grade** (about 7–9 years old) who are learning the multiplication and division tables.
- **Parents**, who can follow the child's progress: completed levels, stars, points and test results.
- **Teachers**, who can use the app as extra practice or for contests between the pupils in their class.

The app uses the terms from the textbook: dividend, divisor, quotient, "3 times smaller", "3 less", checking by multiplication. In English, all the exercises, explanations and word problems are translated.

What the app looks like

The child moves around a **map of islands**. Each island is a lesson or a chapter. The islands unlock one by one: a new island opens after the previous one is finished.

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Home page
├── 🏁 Contest Arena              – contests with other children
└── 👛 Learning Island
    ├── 🧩 Puzzle Island          – 14 islands, from easy to advanced
    ├── 📘 Multiplication table   – the textbook lessons: multiplying by 2 … by 9
    ├── ➗ Division table         – division and the lessons "dividing by 2 … by 9"
    ├── 🎓 National Assessment    – the official 2025 and 2026 tests + review
    ├── 🖍️ Unknown terms          – exercise 46: a + 34 = 62
    ├── ➖ Column subtraction     – 2 digits, with a riddle: discover the animal
    ├── ➕ Column addition        – 2 digits, with a riddle: discover the animal
    ├── 🔢 3-digit subtraction    – 258 – 137, in a column
    └── 🧮 3-digit addition       – 648 + 234, in a column
```

Tapping the abacus in the top-left corner always takes the child back to the home page.

What it contains

🧩 Puzzle Island: exercises by difficulty

14 islands, each with a single type of exercise, ordered from easy to hard:

Difficulty	Islands
🟢 Easy	Additions and subtractions (up to 20) · Multiplications up to 5 × 5
🟡 Medium	More, less (money, ages) · Times tables up to 10 × 10 · Multiplication word problems · Calculate (missing factors, double, triple)
🟠 Hard	Two-step problems · Multiplying by 11 and 12 · Practice like in the textbook · Step-by-step problems · Practise and solve
🔴 Advanced	Big multiplications (up to 12 × 12) · Challenges · **The final test**, which gives a grade (FB, B, S, I – Very good, Good, Satisfactory, Insufficient)

📘 Multiplication table: the textbook lessons

One lesson for multiplying by **2, 3, 4, 5, 6, 7, 8 and 9**. Each lesson has:

- the **"📖 Remember"** box, with the rule and the times table, as in the book;
- 4 levels: **Discover, Apply, Practice, Solve problems**, with the exercises and problems from the textbook.

➗ Division table

- **Division**, with 8 levels:
  - division as repeated subtraction;
  - division as the inverse of multiplication;
  - the dividend, the divisor and the quotient;
  - dividing by 10;
  - special cases: by 1, by itself, dividing 0, dividing by 0 makes no sense;
  - checking multiplication and division.
- **Dividing by 2, 3, 4, 5, 6, 7, 8 and 9**: one island for each, with "Remember" and the 4 levels. Each island only uses division by its own number.

🖍️ Find the unknown terms (exercise 46)

- **12 exercises** in which the child finds the missing number: a + 34 = 62, b − 68 = 19, 45 − d = 26 and others.
- The exercises with brackets are solved **step by step**: first the brackets, then the unknown term.
- For each exercise, the child sees the **rule** ("one term = the sum − the other term") and the **check**.
- **No timer**: the child works at their own pace and can skip an exercise.

✍️ Column addition and subtraction

Four islands with calculations written **vertically, as in an exercise book**, on squared paper: units under units, tens under tens, hundreds under hundreds.

Island	What it contains
➖ Column subtraction	subtraction with 2-digit numbers, with borrowing (73 − 37)
➕ Column addition	addition with 2-digit numbers, with carrying (39 + 15)
🔢 3-digit subtraction	subtraction with 3-digit numbers (258 − 137, 478 − 76)
🧮 3-digit addition	addition with 3-digit numbers (648 + 234, 808 + 8)

- **The child types the result**, digit by digit, in the boxes under the line, instead of choosing it from options. As on paper, they start with the units (the box on the right) and the cursor moves left by itself. On a phone, the number keyboard opens.
- After "✔ Check", the boxes turn **green** (correct) or **red** (wrong).
- **" 💡 How to solve it?"** explains column by column: "Units: we can't do 3 − 7, so we borrow one ten: 13 − 7 = 6…", with borrowing, carrying and the check.
- **Animal riddle** (in 2-digit addition and subtraction, like exercise 8 in the textbook): each calculation has a letter under it. At the end, the child puts the results in **increasing or decreasing order** and reads the letters, which spell the name of an animal (PELICAN, HAMSTER, PANDA, ZEBRA, KOALA…). There is also a clue, for example "a black-and-white bear that eats bamboo".
- **The first round** has the exercises from the textbook. **In every new round**, the game makes new calculations and picks another animal, so the exercises do not repeat.
- **No timer**, as in exercise 46.

🎓 National Assessment (2nd grade)

- **The 2026 test**, "Pets": 18 items.
- **The 2025 test**, "Habitats": 12 items.
- **The 2025 test**, "Animals from the South Pole and the North Pole": 12 items.
- **Review problems**: bears, trees, squirrels, a train, posters.

The tests reproduce **all the items** from the official tests, in the same order and with the same scoring (100 points). The pictures from the test (tables, posters, geometric shapes, sequences) are redrawn in the app. The items where, on paper, the pupil writes out the solution are split **into steps**, with answer choices.

In the tests:

- **there is no timer**;
- the child can **skip an item** and come back to it later, from the list of unsolved items;
- at the end, the child gets a **score out of 100**.

How to play

- **A round** has 10 questions, with 4 answer choices each. In the National Assessment tests, the round has all the items of the test. In the column calculations, the result is typed into boxes.
- **The time** is 30 seconds per question. The National Assessment tests, exercise 46 and the column calculations have no time limit.
- **Lives:** the child starts with 3 lives and can have at most 5. A mistake costs a life, and 5 correct answers in a row give one back. If the child runs out of lives, they can "buy" one for 30 points.
- **Points:** 10 points for each correct answer, plus a "combo" bonus for several correct answers in a row.

- **Stars:** at the end of the round the child gets 1–3 ⭐, depending on the number of correct answers. The map keeps the best result.
- **“ 💡 How to solve it?”** shows the explanation for each question.
- **Long problems are solved step by step.** For example: first how many adult bears there are, then how many bears in total, then how many animals.
- **Pictures:** some questions have drawings, for example groups of flowers, chicks in nests or palm trees.
- **No repeats:** within a round, the same calculation is never repeated, even if it is written differently.
- **Changing the language** works even in the middle of a round: the question on the screen switches to the other language straight away.

The social side

- 🏆 **Leaderboard:**
  - “My scores”: the best score on each level, grouped by island;
  - “All players”: the total score of all the children;
  - “My contests”: the history of contests.
- 🏁 **Contest Arena:**
  - real-time contests with 2–10 children and 10 questions of 35 seconds each, on the chosen level;
  - the winner is whoever has the most correct answers (in a tie, the fastest);
  - point prizes: 🥇 +50, 🥈 +30, 🥉 +10;
  - a table of contents by island and a 🔔 notice when another child has created a contest.
- 💬 **Chat** between players:
  - a general room and private messages, with emoji, replies to a message and notifications;
  - **links** in messages open with one click;
  - **videos** (YouTube, Vimeo or video files) can be watched right in the chat.
- 📞 **Voice calls and** 📹 **video calls**, as on WhatsApp:
  - from the top of the conversation, choose a voice call (sound only) or a video call (with the camera);
  - the camera can be turned on or off during the call, and the microphone can be muted;
  - the conversation keeps a call history (duration, missed, declined), with a “Call back” button.
- **Profile:** name, avatar (photo), country (flag) and an account with email and password, with a password reset option.

How to use it

1. Open the app in a browser (on a computer or a phone).
2. Choose a name. You can play without an account, but with an email and password account your progress is kept on any device.
3. On the home page, choose the **Learning Island** to learn and practise, or the **Contest Arena** to compete with other children.
4. Choose the language in the top bar: RO **RO** or GB **EN**.

The home page also has two more buttons:

- 📲 **Install**: installs the app on a phone or computer so it opens like a regular app (on an iPhone a guide appears: “Share” → “Add to Home Screen”);

- 📄 **About the game**: this document, in the chosen language (Romanian or English).